Nishal Stanislaus Silva, *PhD*

ML Research Engineer | Real-Time & Edge Inference | Audio, Sensor & Time-Series

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Professional Summary

Machine learning research engineer specializing in real-time and edge audio inference under hard latency budgets. Owned the full pipeline for a published streaming audio pattern detector – acquisition, DSP/feature engineering, training, and edge deployment – reaching 14 ms inference on a Raspberry Pi 4 at F1 0.76 and 31.4% CPU, a 74x speed-up over the 1038 ms DTW baseline it replaced (JAES 2026). This is direct evidence of shipping voice/audio-adjacent ML applications against latency and reliability constraints comparable to automotive-grade requirements, backed by C++17 production systems engineering, embedded Linux deployment (Elk Audio OS, JUCE, VST, OSC), and a research discipline of frozen-protocol benchmarking and ablation studies to quantify accuracy/latency/CPU trade-offs.

Core Competencies

Voice/Audio ML on Embedded Systems: DSP feature extraction (STFT, MFCC, LFCC, PLP, GFCC, S-transform, CQT), onset/beat tracking, YIN pitch estimation, real-time audio-model deployment on Elk Audio OS with JUCE, VST, and OSC integration, embedded/edge inference on Raspberry Pi and ARM CPUs under sub-30 ms latency budgets, benchmark and ablation design for accuracy/latency/CPU trade-offs. **Systems & ML:** C++17 (expert, real-time engines), Python (TensorFlow, Keras, PyTorch, scikit-learn), embedded Linux, sensor/IMU fusion, time-series and sequence modelling (RNN/LSTM/GRU), computer vision (OpenCV) for industrial inspection.

Technical Stack

C++17, Python, TensorFlow, Keras, PyTorch, scikit-learn, Elk Audio OS, JUCE, VST, OSC, Librosa, SciPy, FFTW, libsndfile, Raspberry Pi, ARM, embedded Linux, OpenCV, SQL, AWS, Docker, CI/CD, ETL pipelines, Git, pytest.

Education

PhD, Information Engineering and Computer Science – University of Trento, Italy (defended Jan 2025)
MSc, Telecommunication and Electronic Engineering – Sheffield Hallam University, UK (2018)
BEng (Hons), Electronic Engineering – Sheffield Hallam University, UK (2014)

Research and Work Experience

Independent Researcher | *Self-directed, Toronto, Canada* *Jan 2026 - Present*

- Two peer-reviewed papers accepted for publication: Smart Drums (JAES, in press) and MIRaaS (IEEE IS2, accepted), extending the real-time pattern-detection and edge-offload work from the PhD.
- Maintain Nebula, an open-source C++17 audio-feature-extraction library with per-feature ARM/Raspberry Pi latency benchmarks (13 GitHub stars).
- Continue independent benchmark and edge-inference work on audio and time-series ML, using AI-assisted development workflows including Claude Code.

Postdoctoral Researcher | *University of Trento, Trento, Italy* *Jan 2025 - Dec 2025*

- Owned the full ML pipeline for streaming audio and IMU/gesture data on the EU Horizon EIC Pathfinder project MUSMET: acquisition, DSP/feature engineering, training and evaluation, edge deployment (Raspberry Pi 4, Elk Audio OS, VST, OSC), and monitoring.
- Published a real-time audio pattern detector achieving 14 ms inference on Raspberry Pi 4 at F1 0.76 and 31.4% CPU, 74x faster than the 1038 ms DTW baseline it replaced (JAES 2026).

- Designed frozen-protocol benchmark suites and ablations quantifying accuracy/latency/CPU trade-offs across model variants.
- Built MIRaaS, a UDP-based edge-to-server offload architecture with a latency-characterization study accepted at IEEE IS2 2026.

Visiting Researcher | *McGill University (IDMIL / CIRMMT), Montreal, Canada* *May 2024 - Aug 2024*

- Built a self-powered smart electric guitar combining a BNO055 IMU with a HiFiBerry audio codec on a Raspberry Pi 4.
- Fused IMU and audio streams via a hierarchical finite-state machine, reaching F1 0.78 on a 500-event corpus (IEEE IS2 2025).

Visiting Researcher | *University of Visual and Performing Arts, Colombo, Sri Lanka* *Jan 2024 - Mar 2024*

- Conducted a human-centred evaluation study of musicians' perception of smart instruments with embedded pattern detection (IEEE I3DA 2025).

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka* *Aug 2020 - Feb 2021*

- Built ML and statistical pipelines for financial forensics, compliance monitoring, and anomaly detection on large structured enterprise data.

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka* *Jan 2015 - Jul 2020*

- Delivered end-to-end computer-vision and IoT systems for manufacturing at scale across a 5.5-year tenure, with operational efficiency gains of up to 300%.
- Ran real-time IoT data ETL across production sites and deployed C++/Python inference on embedded hardware; introduced CI/CD and code review practices and mentored engineers.
- Automated multilingual care-label quality control using OpenCV with conveyor and PLC integration for automatic reject, cutting inspection time by 99.5%.

Publications

Silva, N. & Turchet, L. (2026). Real-Time Audio Pattern Detection for Smart Musical Instruments. *Journal of the Audio Engineering Society*, 74(3). DOI 10.17743/jaes.2022.0250. Silva, N. & Turchet, L. (2026, in press). Smart Drums. *Journal of the Audio Engineering Society*. Silva, N. & Turchet, L. (2026, accepted). MIRaaS. *IEEE International Symposium on the Internet of Sounds (IS2)*.

Highlights

MIDI Innovation Awards 2023 Finalist (Prototypes & Non-Commercial Software). GMOA recognition (Government Medical Officers' Association, Sri Lanka, 2020) for a COVID-19 remote vitals monitoring deployment. Fluent in English (native); basic, actively improving French (approximately A2).